

Student Project Proposal

MolParetoBO: Surrogate-Assisted Multi-Objective Molecular Optimization

Master area	AI for Drug Discovery; molecular optimization; cheminformatics; active learning; toxicity prediction
Project type	Computational project combining RDKit, machine learning and Pareto optimization
Duration	2-3 months
Supervision	One tutor or co-tutor; weekly or bi-weekly progress meetings
Recommended level	Python, machine learning and cheminformatics basics; familiarity with RDKit is useful
Main output	Documented workflow/notebook, short technical report and final presentation
Possible data	Small molecular dataset with SMILES and property/toxicity scores; public benchmark data or tutor-provided toy oracle

Short description. The project asks the student to implement a simplified MolParetoBO workflow for optimizing molecules against multiple endpoints using molecular mutations, validity filters, surrogate models and Pareto-based candidate selection.

1. Background and rationale

Drug discovery rarely involves optimization of a single property. A candidate molecule must balance potency, toxicity, solubility, permeability, synthetic accessibility and other ADMET-related endpoints. These objectives are often conflicting, so a single weighted score can hide important trade-offs.

Pareto optimization provides a natural framework for this problem because it identifies molecules that cannot be improved in one objective without worsening another. However, evaluating every molecule with expensive oracles is not feasible.

MolParetoBO addresses this challenge by combining molecular mutations, fast surrogate models, active learning and multi-objective selection. For a student project, a simplified prototype can capture the essential logic without implementing the full research-grade framework.

2. Project objective

The objective of the project is to build a small-scale surrogate-assisted multi-objective molecular optimization workflow. Starting from one or more seed molecules, the workflow should generate valid candidates, evaluate or predict multiple endpoints, update a Pareto front and select promising molecules for further evaluation.

The project should be implemented as a clear proof of concept. Complex components such as full NSGA-III, qEHVI or graph neural networks may be simplified depending on the student background and available time.

3. Specific aims

- Generate molecular variants using RDKit-based mutations, fragment replacement or simple bioisosteric rules.
- Apply validity and drug-likeness filters such as sanitization, molecular weight, QED, PAINS or synthetic accessibility proxies.
- Define two to five endpoints representing toxicity, drug-likeness or other molecular properties.
- Train a surrogate model to approximate endpoint evaluations and, where possible, estimate uncertainty.
- Maintain and visualize an approximate Pareto front of non-dominated molecules.
- Implement an active selection strategy that balances predicted quality, uncertainty and chemical diversity.

4. Proposed methodology

4.1 Molecular generation and filtering

The student will start from seed molecules and generate candidate variants through simple mutation rules. RDKit will be used to validate structures, calculate descriptors and remove chemically implausible or undesirable molecules.

4.2 Endpoint definition

A small set of objectives will be defined. These may include predicted toxicity scores, QED, logP, molecular weight, synthetic accessibility, similarity constraints or tutor-provided oracle values.

4.3 Surrogate modelling

A baseline model such as Random Forest, Gaussian Process, gradient boosting or a simple neural network will be trained on molecular fingerprints and endpoint values. The model will be updated as new molecules are evaluated.

4.4 Pareto search and acquisition

Candidates will be ranked by Pareto dominance and by a simple acquisition score combining predicted performance, uncertainty and novelty. The selected molecules will be evaluated by the oracle or by the predefined scoring function.

4.5 Analysis of results

The student will track hypervolume or simpler Pareto-quality indicators, chemical diversity and convergence behaviour. Final molecules should be discussed as trade-offs rather than as a single best compound.

5. Indicative work plan

Period	Main task	Expected output
Weeks 1-2	Problem setup and seed dataset	SMILES dataset, endpoint definitions and baseline descriptors.
Weeks 3-4	Molecular mutation and filtering	Candidate generation module with validity and property filters.
Weeks 5-6	Surrogate model	Baseline predictive model for multiple endpoints.
Weeks 7-8	Pareto front and acquisition	Non-dominated sorting and candidate selection strategy.
Weeks 9-10	Optimization experiments	Iterative search results, diversity analysis and front visualization.
Weeks 11-12	Final report and presentation	Final workflow, figures, limitations and future extensions.

6. Expected deliverables

- A documented Python notebook or repository implementing the simplified MolParetoBO workflow.
- A candidate-generation and filtering module using molecular representations.
- A surrogate model for two or more molecular endpoints.
- Pareto front visualizations and summary metrics for optimization progress.
- A final technical report and a 10-15 minute presentation.

7. Skills acquired by the student

- Practical cheminformatics with SMILES, fingerprints and molecular descriptors.
- Implementation of multi-objective optimization concepts in drug discovery.
- Use of surrogate models and active learning for expensive evaluations.
- Interpretation of Pareto fronts and trade-offs among molecular properties.
- Critical evaluation of validity, novelty and synthetic feasibility of generated molecules.

8. Suggested software and resources

The project can be implemented with RDKit, pandas, NumPy, scikit-learn, matplotlib and optionally pymoo, BoTorch or GPyTorch for advanced optimization components.

The tutor should define whether the project uses real endpoint data, public benchmark scores or a toy oracle. For most students, a simplified acquisition strategy is preferable to a full qEHVI implementation.

9. Optional extensions

Possible extensions include GP-based uncertainty, NSGA-II or NSGA-III search, qEHVI-style acquisition, GNN surrogate models, or inclusion of potency as an additional objective together with toxicity and drug-likeness.

10. Evaluation criteria

Criterion	What should be assessed
Scientific clarity	The scientific question, assumptions and expected use case are clearly stated.
Technical implementation	The code is readable, documented and reproducible on the selected data.
AI relevance	The project produces representations, models or workflows that are suitable for AI-based analysis.
Interpretability	The student explains what the results mean in a drug discovery or molecular science context.
Critical discussion	The final report discusses limits, uncertainty and possible future extensions.

Note for tutors. The minimal successful version should implement candidate generation, filtering, multi-objective scoring and Pareto visualization. Advanced Bayesian optimization and GNN surrogates are optional extensions.